1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In metastatic castration-resistant prostate cancer (mCRPC) treated with docetaxel, a high stromal-to-tumor ratio (TSR) in pretreatment tumors is associated with reduced therapy_response. This association is mediated by two linked mechanisms: (1) TSR-driven microenvironmental cues promote AR signaling reactivation in tumor cells, and (2) AR signaling upregulates the drug-efflux transporter ABCB1, decreasing intratumoral docetaxel exposure and cytotoxicity. Consequently, TSR-high tumors with evidence of AR reactivation and elevated ABCB1 will exhibit the poorest therapy_response compared with TSR-low tumors. Secondary modifiers (CYP17A1, TUBB3, PTEN, SLCO2B1) may modulate the magnitude of these effects but are not required for the primary mechanism. The primary measurable endpoint to test this hypothesis is therapy_response (objective radiographic/biochemical response to docetaxel).

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	AR, ABCB1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

AR: ard cytotoxic option; why does chemotherapy fail in certain castration-resistant, metastatic tumors? 2) Tier 1 paths used: (A) castration-resistance → AR reactivation (from Node 1 definition) and (B) mCRPC → docetaxel-based therapy (Nodes 2–4).

ABCB1: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CYP17A1	Supporting modifier from knowledge graph
Secondary	TUBB3	Supporting modifier from knowledge graph

Secondary	PTEN	Supporting modifier from knowledge graph
Secondary	SLCO2B1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Clinical correlative study measuring pretreatment TSR in tumor biopsies and associating with documented docetaxel therapy_response;

Prediction 2: longitudinal assessment of AR target gene expression and ABCB1 levels in tumors before/after docetaxel exposure;

Prediction 3: preclinical models (mCRPC cell lines, patient-derived xenografts) engineered or selected for TSR levels to test docetaxel accumulation and cytotoxicity, with AR antagonists or ABCB1 inhibitors to assess causality;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- CYP17A1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- TUBB3: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.